Track 1 Final Methods & Clinical Report

Identification of Causal Compound Heterozygous *BUB1B* Variants in Mosaic Variegated Aneuploidy (MVA1)

TEAM: PROFETAS (DBLAS1) | PROBAND: PROBAND01 | SAGEBIO RARE DISEASE HACKATHON 2026

100.0 / 100	1.000	Rank 1
RANK POINTS	F-MAX METRIC	FULL MATCH THRESHOLD

1. EXECUTIVE SUMMARY & DIAGNOSTIC RESULT

We report the resolution of **PROBAND01** in Track 1 of the SageBio Rare Disease Hackathon 2026. By executing an end-to-end variant prioritization pipeline on Whole Genome Sequencing (WGS) data (~85 GB), team **profetas** identified and validated compound heterozygous pathogenic mutations in the ***BUB1B*** gene (chr15) as the primary cause of Mosaic Variegated Aneuploidy Syndrome 1 (MVA1). The automated pipeline achieved a perfect **100.0 Rank Points** and an **F-max of 1.000** on the evaluation platform.

2. IDENTIFIED CAUSAL VARIANTS IN *BUB1B*

Variant Property	Allele 1 (Truncating / Nonsense)	Allele 2 (Missense / Disruption)
Genomic Coordinate (GRCh38)	`chr15:40209701 T>G`	`chr15:40220612 T>G`
HGVS Transcript Reference	`NM_001211.6:c.2210T>G`	`NM_001211.6:c.3006T>G`
Protein Consequence	`p.Leu737Ter` (Nonsense)	`p.Asn1002Lys` (Missense)
ACMG Criteria	Pathogenic (PVS1, PM2, PP3)	Likely Pathogenic (PM2, PM3, PP3)

Compound Heterozygosity Mechanics in PROBAND01

Biallelic functional inactivation occurs through the combination of `p.Leu737Ter` (leading to protein truncation or nonsense-mediated decay) and `p.Asn1002Lys` (disrupting the key C-terminal kinase domain and BUB3 interaction). Trans-phase inheritance confirms complete functional loss of BUBR1, precipitating MVA1 phenotype onset.

3. PATHOPHYSIOLOGY & LITERATURE EVIDENCE MAPPING

Mosaic Variegated Aneuploidy Syndrome 1 (MVA1, OMIM #257300) is an autosomal recessive genomic instability syndrome characterized by constitutional aneuploidies in over 25% of somatic cells. Affected patients present with severe intrauterine growth restriction (IUGR), microcephaly, developmental delays, cerebellar hypoplasia, and a predisposition to embryonal childhood tumors (rhabdomyosarcoma, Wilms tumor).

The *BUB1B* gene encodes **BUBR1**, a central protein kinase operating within the **Spindle Assembly Checkpoint (SAC)**. BUBR1 ensures proper chromosome segregation by forming the Mitotic Checkpoint Complex (MCC) to inhibit premature onset of anaphase via APC/C inhibition.

4. LITERATURE EVIDENCE MAPPING (CONSENSUS AI & PUBMED)

- **Matsuura et al. (2006) / PubMed ID 16826528:** First demonstrated that bi-allelic mutations in *BUB1B* lead to SAC impairment and severe constitutional mosaic aneuploidy in humans. Automated mapping via **Consensus AI** confirmed high literature consensus (100% agreement across 18 publications) for *BUB1B* biallelic loss in MVA1.
- **Garcia-Castillo et al. (2008):** Confirmed that compound heterozygous truncating and missense variants in the kinase domain lead to severe mitotic checkpoint insufficiency and microcephaly.

5. PIPELINE ARCHITECTURE & AI DISCLOSURES

Genomic Pipeline Workflow: Whole genome VCFs (~85 GB) were processed locally using `pysam` and `scikit-allel`. High-confidence variants ($DP \geq 20$, $GQ \geq 60$, gnomAD $AF < 0.001$) were annotated with Ensembl VEP and prioritized using LightGBM trained on ACMG guidelines and HPO phenotypic features (microcephaly: `HP:0000252`).

AI Collaboration Disclosure: Advanced literature review and report structuring were assisted by Google Gemini API models (commercial enterprise terms, no data training or retention).